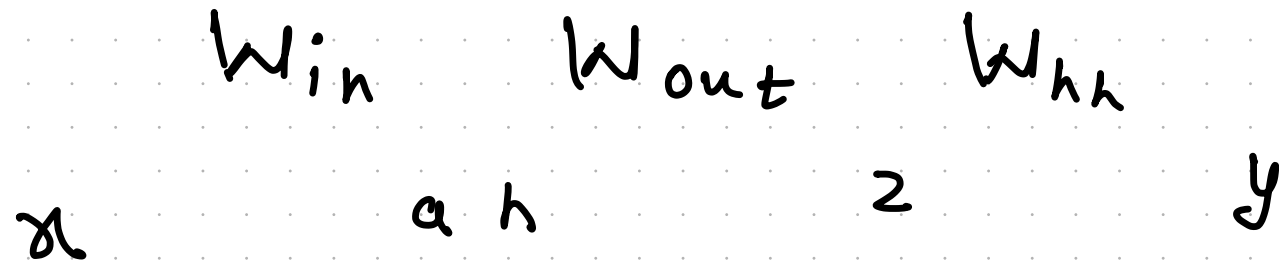


RNNs can be understood as extension of FFNN

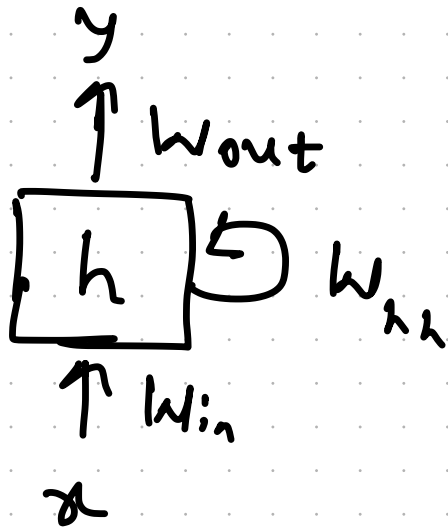
- Elman's RNN : 1990



- Backprop through time
- Vanishing and Exploding gradient
- Remedies for gradient problems
- RNN configurations

Time

Unrolling



Let us compute h_4

BPTT

$$E(\theta) = \sum_{i=1}^t L_i$$

or just L_t

$$\frac{\partial E}{\partial W_{out}}$$



Same
as
FFNN

$$\frac{\partial E}{\partial W_{in}}$$



Same
as
FFNN

$$\frac{\partial E}{\partial W_{hh}} = \frac{\partial E}{\partial y_t} \cdot \frac{\partial y_t}{\partial h_t} \cdot \frac{\partial h_t}{\partial W_{hh}}$$



Gradient
problems

Bidirectional RNN

Stacked RNN

Attention Mechanism

- Why fight with gradient problems if we can remember the whole history
- Luong attention (2015)
 - Dot product: $\vec{h}_t^T \vec{h}_i$
 - Learnable similarity: $\vec{h}_t^T W_a \vec{h}_i$
 - Concat: $V_a^T \cdot \tanh(W_a [\vec{h}_t; \vec{h}_i])$
 - Convert scores to probabilities
 - Content vector: weighted sum of history: c_t
 - New hidden state: $\tanh(W_k [c_t; \vec{h}_t])$
 - Global vs Local

Attention Mechanism

- Bahdanau attention (2015)

- $$e_{t,i} = v_a^T \tanh(W_s \cdot s_{t-1} + W_h h_i)$$

- Convert all $e_{t,i}$ s to probabilities

- Context vector is weighted sum of history: c_t

- $$s_t = f(s_{t-1}, y_{t-1}, c_t)$$

- Bahdanu directly changes decoder side hidden state

- Attention alignment visualization : Attention Maps

The Deep Learning . Timeline

- 1943: Idea of artificial neuron, McCulloch
- 1958: Perceptron with step function, Rosenblatt
- 1974: Backprop, Werbos but ignored by the field
- 1980: Concept of CNN, No backprop, Fukushima
- 1986: Backprop as we know it, Hinton
- 1990: RNN, Elman
- 1992: BiRNN, Paliwal
- 1998: CNN, LeCun
- 2001: Stacked RNN
- 2014: Encoder Decoder architecture, Sutskever
- 2015: Attention, Bahdanau
- 2017: Transformer, Vaswani

A few missed points

- Teacher forcing during training
- $\langle \text{sos} \rangle$ token at beginning of decoding
- Embedding matrix E

$E \cdot x$
↑
One hot encoding vector

↑
Learnt as part of the training

We are selecting a column of E

- AI by Hand: Tom Yeh
↓ ↓
Website Book

Self Attention Before Transformer

- Lin attention, 2017

- RNN encoder generates $h_1, h_2 \dots h_T$
- H : Each row is a hidden state ($T \times d$)
- A : $r \times T$
- $A = \text{softmax}(W_{s2} \cdot \tanh(W_{s1} \cdot H^T))$
- Each row of A sums to one
- W_{s1} : $d \times d$ W_{s2} : $r \times d$
- $M = A H$ ($r \times d$)
- Each row of M is a semantic summary of the input.
- How to use M :
 - Flatten by concatenating
 - Pooling: Min, Max, Average
 - Matrix